

Schedule:

Day	Topics
Week 1	
Day 1	AM class: Sets and real numbers, absolute value and distance, algebraic and fractional expressions, exponentials and radicals. PM class: Inequalities, the coordinate plane, graphs of equations, circles, lines, equations.
Day 2	AM class: Functions, graphs of functions, getting information from the graph of a function. PM class: Transformations of functions, combining functions, one-to-one, onto and inverses.
Week 2	
Day 1	AM class: Quadratic functions, polynomials, the Remainder and Factor theorems, the Intermediate Value Theorem for Polynomials and the Rational Zeros Theorem. PM class: Graphs of polynomial functions, rational functions, polynomial and rational inequalities.
Day 2	AM class: Exponential functions, the natural exponential function, logarithmic functions. PM class: Laws of logarithms, exponential and logarithmic equations, fitting exponential and power curves to data.
Week 3	
Day 1	AM class: Angle measure, trigonometry of right triangles, trigonometric functions of angles. PM class: Math modelling hands-on activities.
Day 2	AM class: Trigonometric graphs, inverse trigonometric functions and their graphs. PM class: Trigonometric identities, addition and subtraction formulas, double-angle formulas.
Week 4	
Day 1	AM class: Polar coordinates, graphs of polar equations, plane curves and parametric equations. PM class: Vectors in two dimensions, the dot product, three-dimensional coordinate geometry.
Day 2	AM class: Vectors in three dimensions, the cross product, equations of lines and planes. PM class: Sequences and summation notations, arithmetic sequences, geometric sequences, the Binomial Theorem.
Week 5	
Day 1	AM class: Limits of sequences, limits of functions at infinity. PM class: Limits of functions at a finite value, one sided limits of functions, continuity.
Day 2	AM class: Tangent lines and derivatives, rules of differentiation. PM class: Derivatives of trigonometric, exponential and logarithmic functions, higher order derivatives.
Week 6	
Day 1	AM class: Graphs and derivatives, optimization and derivatives. PM class: Antiderivatives and indefinite integrals, properties of indefinite integrals, rules of integration, signed areas and definite integrals, properties of definite integrals.
Day 2	AM class: The Fundamental Theorem of Calculus, integration by substitution. PM class: Integration by partial fractions, integration by parts.